AHAMMAD NAFIZ

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TECHNICAL SKILLS

- **Languages:** Python, C++, SQL
- **Libraries:** NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, OpenCV, Matplotlib, Seaborn
- **Frameworks:** LangChain, LlamaIndex, Hugging Face, CrewAI, AutoGen, FastAPI, Streamlit
- **Tools:** Git, Docker, Linux, VIM, VS Code, Jupyter, Google Colab, MLflow
- **Databases:** MySQL, Vector Databases (Chroma, FAISS, Pinecone, Weaviate)
- **Domains:** Deep Learning, Computer Vision, NLP, LLMs, GenAI, RAG, MLOps

PROJECTS

Predicta – ML Pipeline Platform **2024 – Present**

- Built end-to-end ML workflow automation for EDA, feature engineering, and model selection
- **Award:** Champion, CSE Project Show Spring 2024 at UIU
- GitHub | *Self-Project*

Analyzia – AI CSV Analysis Platform **2025**

- Designed chat-based CSV explorer using LangChain + Agent + RAG
- Enabled query-to-visual pipeline with real-time validation
- GitHub | *Self-Project*

YOLOv12 Bangladeshi Food Detection **2024**

- Trained YOLOv12m (19.6M params) on 3988-image, 3-class dataset (peyaju, singara, beguni)
- Achieved **87.7% mAP**@0.5, 151 FPS inference, and **82.7% F1** Score
- GitHub | *Self-Project*

Personal Knowledge Assistant – Multi-Agent AI **2025**

- Built web-integrated multi-agent RAG system with semantic search and routing
- Applied document chunking, vector store retrieval, and LLM orchestration
- GitHub | *Self-Project*

EduGen – Curriculum-Aligned LLM Generator **2025**

- Developed an automated system generating curriculum-aligned animated educational content using large language models.
- Created multi-modal outputs combining clear explanations with **3Blue1Brown-style animations** scripted via the `Manim` library for engaging, concept-driven learning.
- GitHub | *(Self-Project)*

EXPERIENCE

Head of Research & Development (UIU Data Science Club) **Mar 2025 – Present**

- Led AI/ML research, student projects, and workshops on transformers and deployment.
- Organized industry collaborations and hackathons to boost practical skills.

Teaching Assistant (United International University) **Jun 2024 – Jul 2024**

- Taught OOP in Python to 40+ students, emphasizing debugging and advanced programming concepts
- Boosted average student performance by **25%** via structured mentoring and personalized guidance

Illustrator & Brand Designer (Freelance) **Aug 2023 – Present**

- Delivered 20+ freelance branding projects using Adobe Suite with **95% client satisfaction rate**

ACHIEVEMENTS

- **Best Prototype Award** – Blockchain Olympiad, UIU CSE Fest 2025 (Team Auctra, 20,000 prize)
- **Champion** – UIU CSE Project Show Spring 2024 for Predicta (Data Science automation tool)
- **1st Runners Up** – Data Innovators Challenge 2024 (Sales analytics using Python + Power BI)

EDUCATION

United International University, Dhaka **Sep 2023 – Present** **B.Sc. in Data Science**